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FROM DUMPSITES TO DECISIONS

An Integrated Life-Cycle, System-Dynamics, Artificial-Intelligence, Multi-Criteria, and Reinforcement-Learning Framework for Sustainable Municipal Solid Waste Management in India

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01 INTRODUCTION

India generates **≈170,000 t of municipal waste per day**; processing reached **54%** in 2021–22, while **2,479 legacy dumpsites** hold ≈250 Mt [3, 6]; global generation heads to 3.8 Bt/yr by 2050 [1]. The tools that could steer the circular transition are mature but **siloed**. This work couples six of them into one framework, on one validated panel of 34 Indian cities (2019–2023).

02 OBJECTIVES

- ✓ Couple LCA, SD, AI forecasting, fuzzy TOPSIS, RL and GP/ARDL through one shared data schema.
- ✓ Score every layer against naive baselines, with uncertainty [57, 58].
- ✓ Audit the data foundation itself and disclose its limits first.
- ✓ Deliver 2035 pathways and allocations a city can act on.

03 METHODOLOGY — LAYERS & DATAFLOW

Figure 1 traces the dataflow: each layer consumes the artifact above it and returns a typed table to the shared schema.

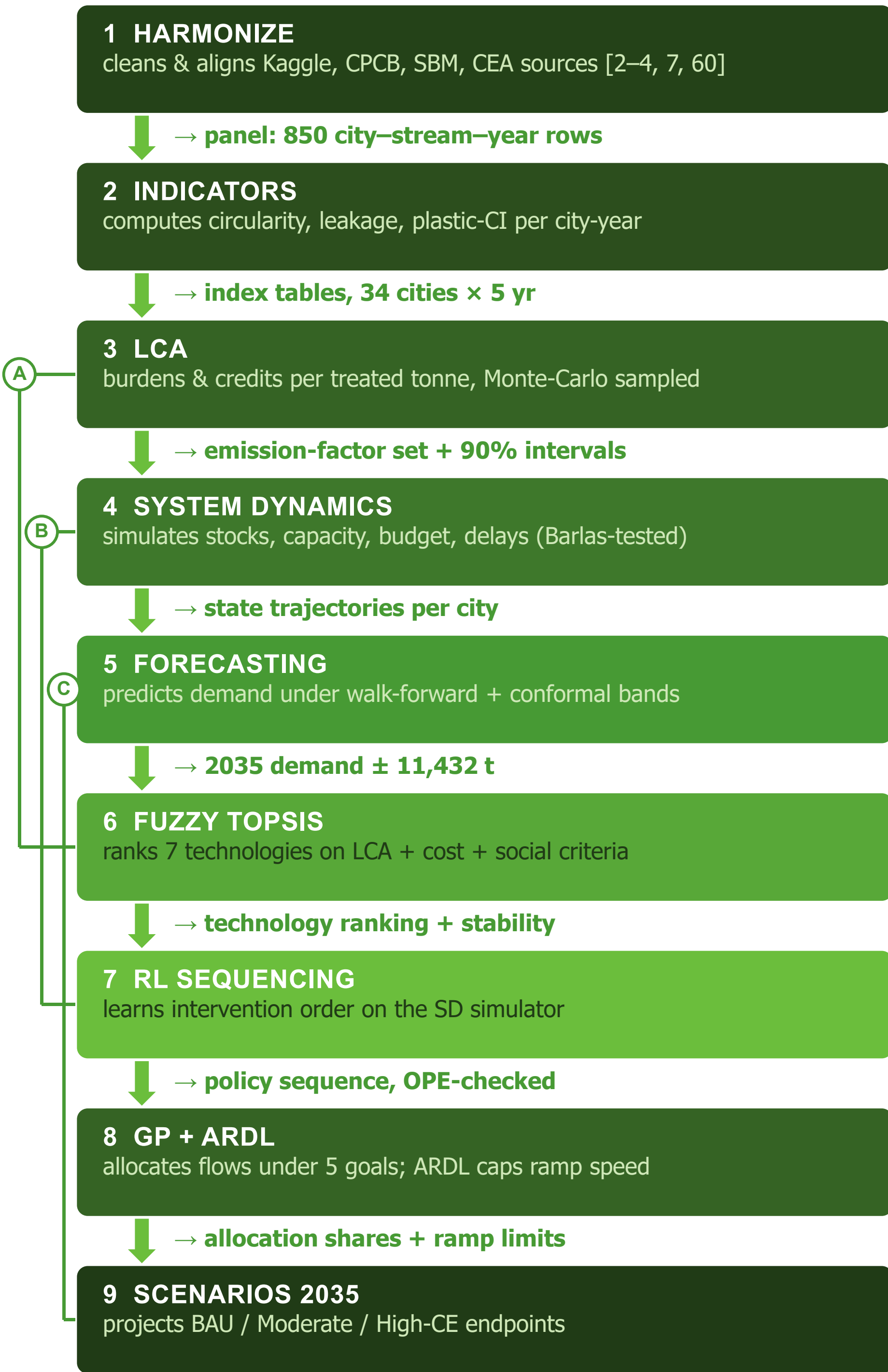


Figure 1. What each layer does and hands on. Side channels: A LCA emission factors → TOPSIS criteria · B SD state = RL environment · C demand forecast → scenario input.

DATA HONESTY — the panel's quantity column behaves as uninformative noise ($|r| < 0.05$, no persistence); forecasts are therefore reported only as skill vs naive baselines, and conclusions rest on composition, treatment shares, and index dynamics.

04 RESULTS

Every number below is walk-forward, interval-quantified, and benchmarked (Figs. 2–6, Table 1).

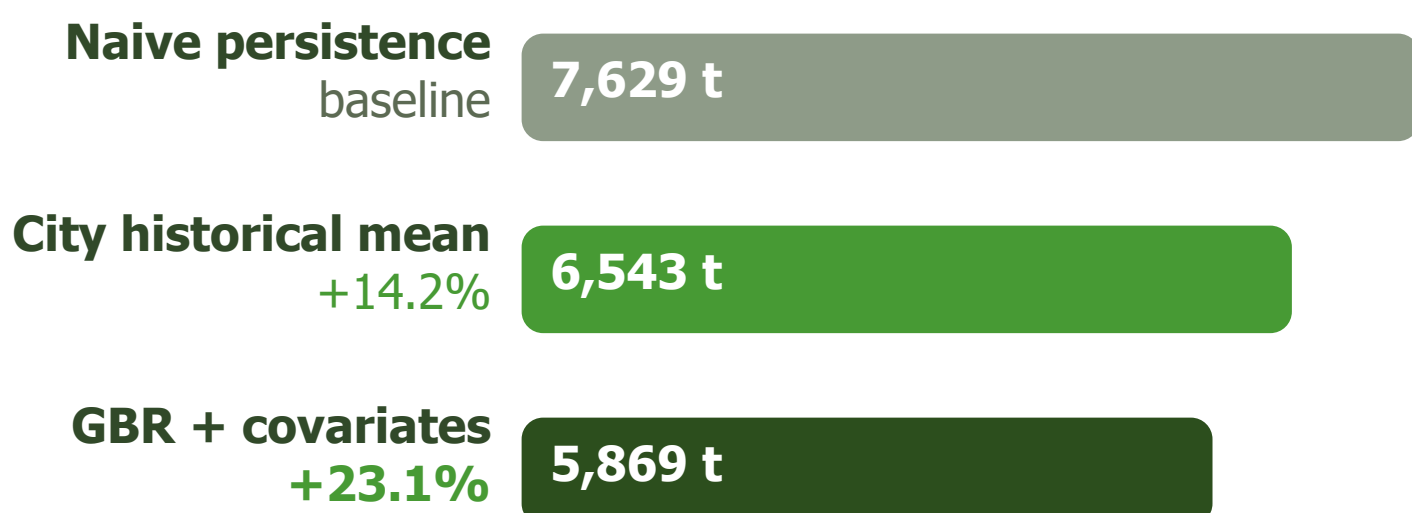


Figure 2. Walk-forward MAE (t/yr), final held-out year, 34 cities; conformal 90% band ±11,432 t at 91.2% coverage.

PARAMETER	BASELINE	THIS WORK	Δ
Forecast MAE (t/yr)	7,629	5,869	–23.1%
90% interval coverage	0.900*	0.912	+0.012
Circularity index 2035	0.432	0.561	≈ +30%
GHG vs BAU 2035	0	–7.8% [5.7–10.1]	—
Plastic circularity 2035	—	0.842 [0.796–0.888]	—
Collection route km	31,010	23,570	–24%
OPE error, WIS · FQE	—	0.155 · 0.161	—

Table 1. Headline results vs baselines (*nominal). Preemptive GP attains all five goals with zero adverse deviation.

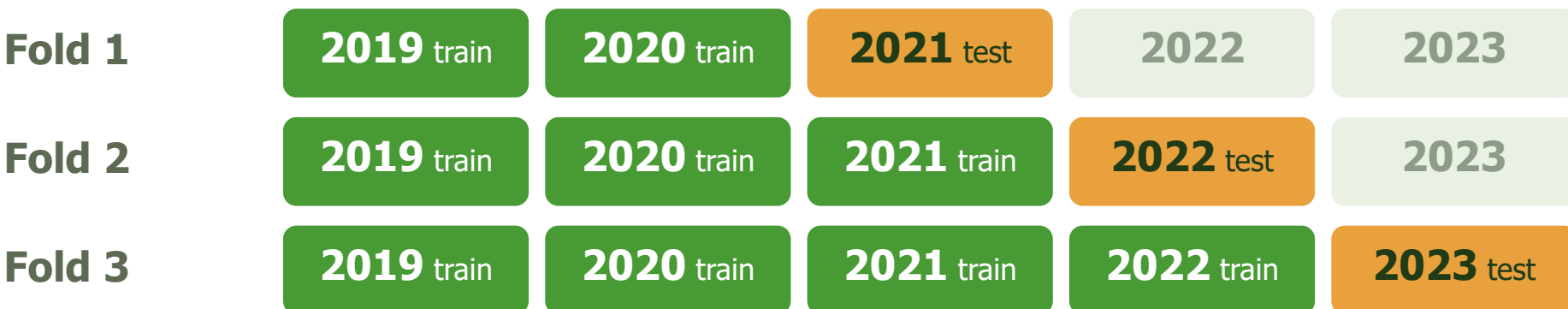


Figure 3. Expanding-window protocol — no temporal leakage [58]; an earlier single-split $R^2 = 0.26$ is reported as an artifact.



Figure 4. High-CE circularity pathway: ARDL-disciplined ramp (schematic between endpoints); 2035 endpoint with 90% interval. Dashed line = 2023 baseline.

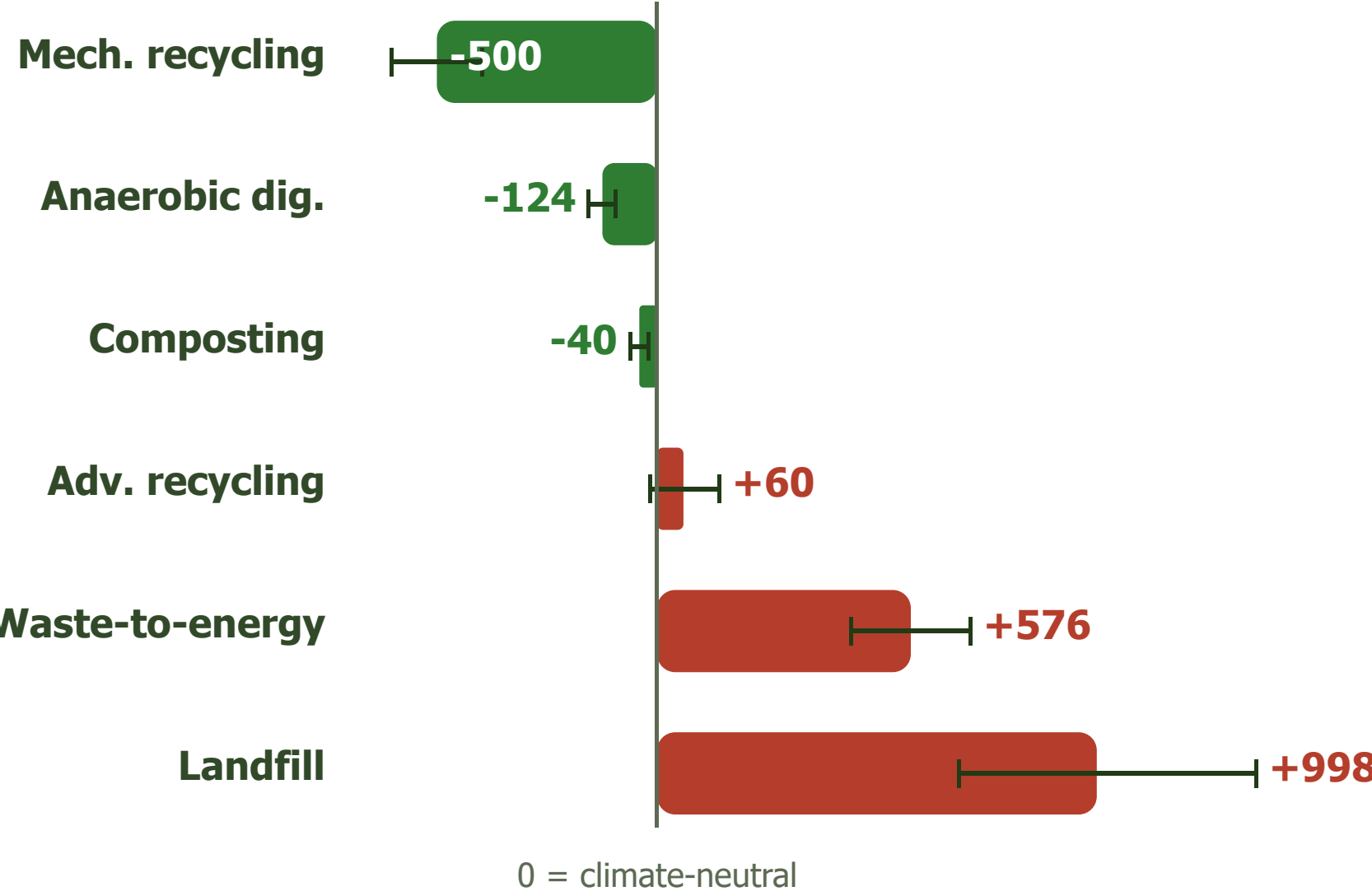


Figure 5. Net GHG, kg CO₂e per tonne — system-wide accounting with material credits; whiskers = 90% MC interval [10–16, 37].

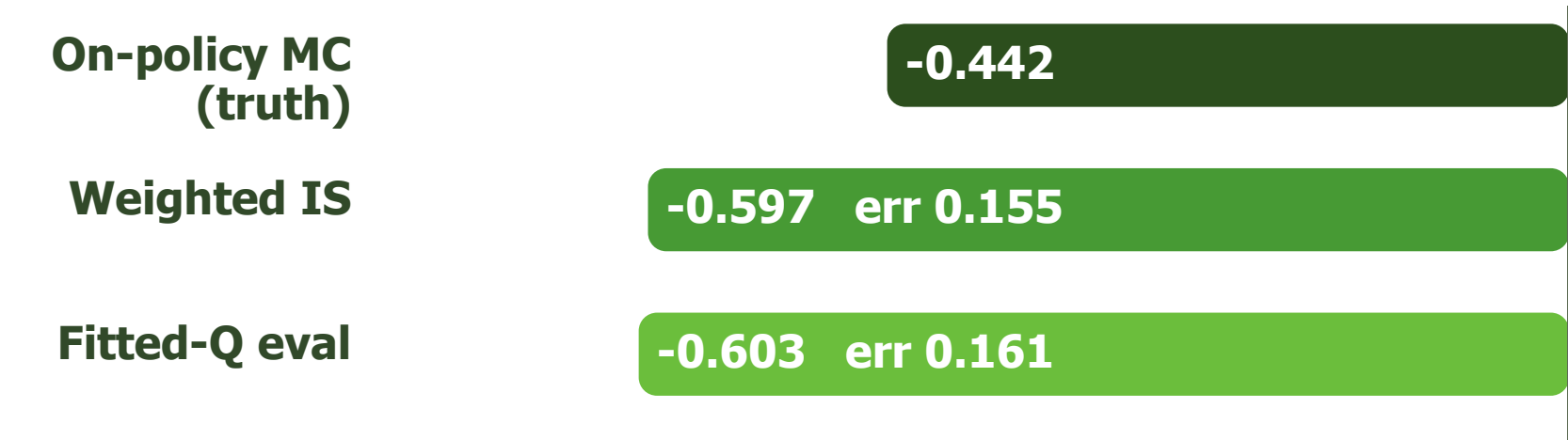


Figure 6. Off-policy evaluation reproduces the RL policy's simulated value — the sim-to-real safeguard [69, 70].

05 DISCUSSION

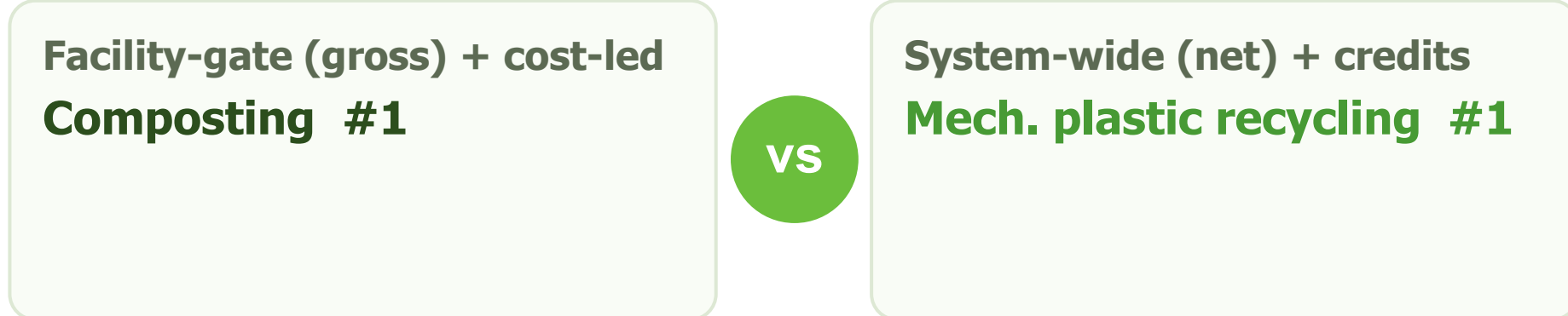


Figure 7. The GHG accounting convention flips the best technology [15, 37, 49] — it is reported as a decision variable.

Stable once fixed: TOPSIS, SAW, WPM, VIKOR agree on the same #1; top rank survives weight perturbation with P 0.96–0.999 (Spearman 0.94–0.98) [49, 51].

Informal workforce: 1.5–2 M waste pickers deliver 20–30% of recovery [26, 52]; livelihood-safe formalization is a criterion, an RL action, and goal G5.

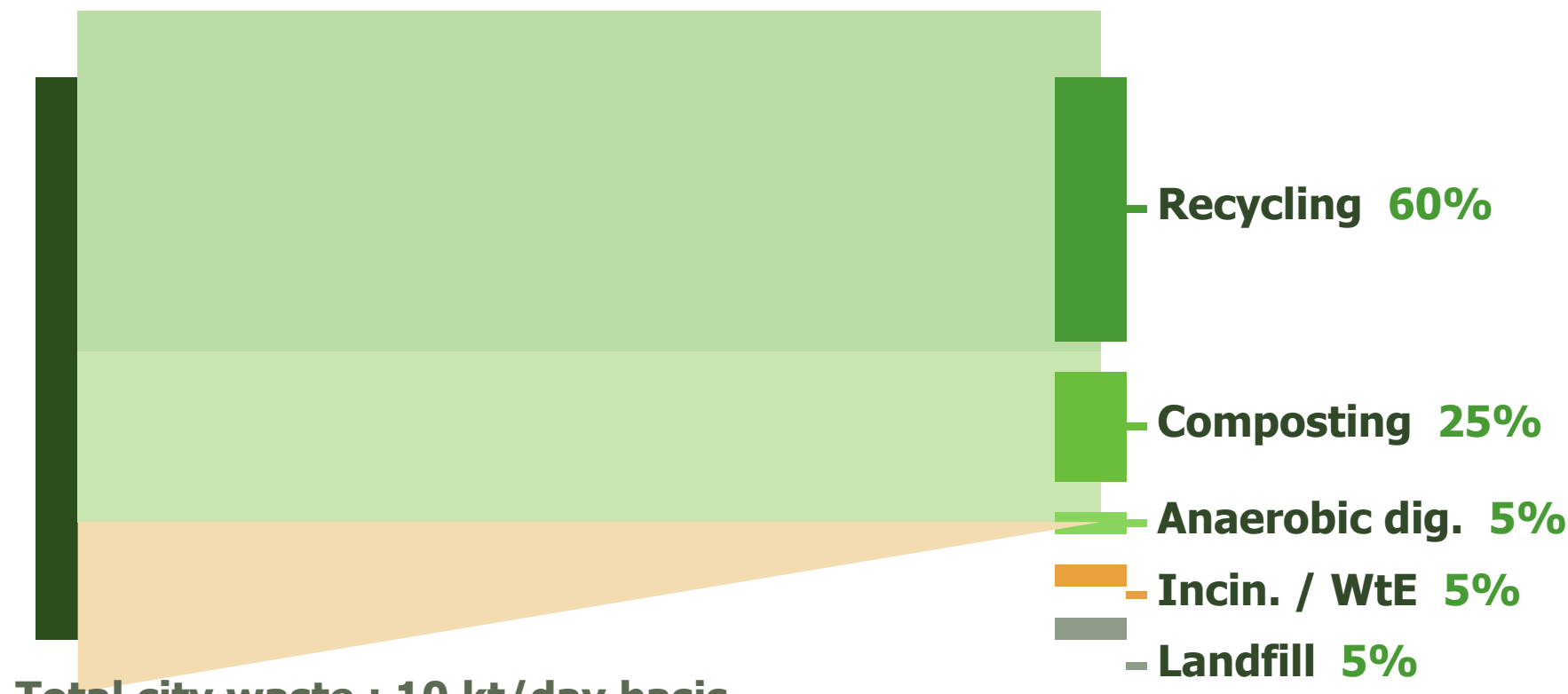


Figure 8. GP-optimized flow allocation (pipeline output, optimized_waste_allocation): recovery routes absorb 90% of flow; landfill drops to 5%.

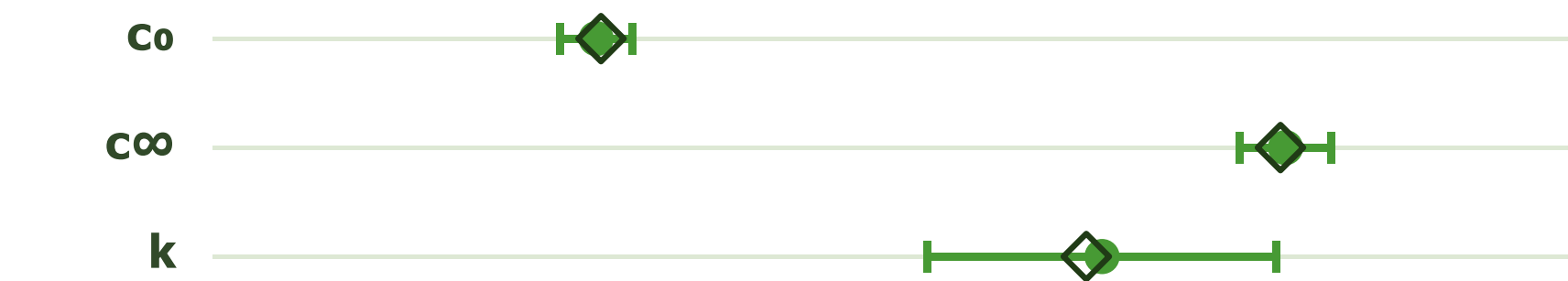


Figure 9. SD identifiability check — synthetic recovery: ◇ true vs ● recovered ± sd; bias < 0.01 on all rate constants [41].

06 CONCLUSIONS

- ▶ Six layers, one benchmark-checked Indian panel — a first for city-level MSW planning.
- ▶ Validation-first: baseline skill, conformal intervals, rank stability, OPE.
- ▶ An honestly audited, fully released data foundation and codebase.

“ A validated circular pathway lifts city circularity from 0.432 to 0.561 by 2035 — and the defensible climate dividend is –7.8%, not the headline 30–45%. — KEY FINDING

07 FUTURE WORK

Primary field data to replace the synthetic quantity column; deep-RL scaling with safe-deployment gates; SD–MFA coupling; more cities; and a live ULB pilot starting from the Indore end-to-end demonstration.

08 REFERENCES

- 1 UNEP & ISWA. Global Waste Management Outlook 2024. wedocs.unep.org
- 2 Kaza, S. et al. What a Waste 2.0. World Bank (2018). hdl: 10986/30317
- 3 CPCB. Annual Report on Solid Waste Management 2021–22 (2023). cpb.nic.in
- 15 Gentil, E. et al. Waste Manag. Res. 27, 696–706 (2009).
- 37 Laurent, A. et al. Waste Manag. 34, 589–606 (2014).
- 41 Barlas, Y. Syst. Dyn. Rev. 12, 183–210 (1996).
- 49 Opricovic, S. & Tzeng, G.-H. Eur. J. Oper. Res. 156, 445–455 (2004).
- 52 Wilson, D. C. et al. Habitat Int. 30, 797–808 (2006).
- 57 Hyndman, R. J. & Koehler, A. B. Int. J. Forecast. 22, 679–688 (2006).
- 58 Bergmeir, C. & Benítez, J. M. Inf. Sci. 191, 192–213 (2012).
- 60 Yadav, K. Waste Management & Recycling in Indian Cities. Kaggle (2024).

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